OPTIMISATION

1. a) The stationary points of the function are the solutions of the equation

$$0 = \nabla f = \begin{bmatrix} x_1 + x_2 - 2 + (r-1)(x_1 - x_2) \\ x_1 + x_2 - 2 - (r-1)(x_1 - x_2) \end{bmatrix}.$$

For any $r \geq 2$ the only solution is $x_{1,\star} = x_{2,\star} = 1$. The Hessian matrix of the function f is

$$\nabla^2 f = \begin{bmatrix} r & 2-r \\ 2-r & r \end{bmatrix}$$

and this is positive definite for all $r \geq 2$. The function f is therefore quadratic and strictly convex: its unique stationary point $X_\star = (1,1)$, is a global minimizer. [4 marks]

b) The gradient iteration for the minimization of f is

$$\begin{aligned} X_{1,k+1} &= X_{1,k} - \alpha_k (X_{1,k} + X_{2,k} - 2 + (r-1)(X_{1,k} - X_{2,k})), \\ X_{2,k+1} &= X_{2,k} - \alpha_k (X_{1,k} + X_{2,k} - 2 - (r-1)(X_{1,k} - X_{2,k})), \end{aligned}$$

where α_k is the line search parameter. [2 marks]

c) Set $r = 100$ as indicated in the text.

i) The gradient of f is parallel to the x_1 axis if its second component is zero. Setting $X_{0,1} = 5$ in $\dfrac{\partial f}{\partial x_2}$ and equating to zero yields

$$0 = \frac{\partial f}{\partial x_2} = 3 + X_{2,0} - 99(5 - X_{2,0}),$$

whence

$$X_{2,0} = \frac{123}{25}.$$

As a result $X_0 = \left(5, \dfrac{123}{25}\right)$. [4 marks]

ii) Replacing X_0 in the gradient iteration one has

$$X_{1,1} = 5 - \alpha_0 \frac{396}{25}, \qquad X_{2,1} = x_{2,0} = \frac{123}{25}.$$

As a result

$$f(X_1(\alpha_0)) = \frac{313632}{25}\alpha_0^2 - \frac{156816}{625}\alpha_0 + \frac{792}{25}.$$

The exact line search parameter is obtained finding the (global) minimizer of $f(X_1(\alpha_0))$. This is achieved, as indicated, for $\alpha_0^* = 1/100$. Finally $X_{1,1} = \dfrac{3026}{625}$, that is $X_1 = \left(\dfrac{3026}{625}, \dfrac{123}{25}\right)$, and

$$\nabla f(X_1) = \begin{bmatrix} 0 \\ -\dfrac{9702}{625} \end{bmatrix},$$

that is the gradient of the function f at X_1 is parallel to the x_2 axis.
[4 marks]

iii) Similarly to the previous point, one has

$$X_{1,2} = X_{1,1} = \frac{3026}{625}, \qquad X_{2,2} = \frac{123}{25} - \alpha_1 \frac{9702}{625}$$

and

$$f(X_2(\alpha_1)) = \frac{188257608}{15625}\alpha_1^2 - \frac{94128804}{390625}\alpha_1 + \frac{475398}{15625}.$$

Minimizing $f(X_2(\alpha_1))$ yields the exact line search parameter, which is again $\alpha_1^* = 1/100$. Finally $X_{2,2} = \frac{148899}{31250}$, that is $X_2 = \left(\frac{3026}{25}, \frac{148899}{31250}\right)$, and

$$\nabla f(X_2) = \begin{bmatrix} -\frac{237699}{15625} \\ 0 \end{bmatrix},$$

that is the gradient of the function f at X_2 is parallel to the x_1 axis.

[4 marks]

iv) The distances between X_0 and $X_\star$ and between X_0 and X_2 are

$$\|X_0 - X_\star\| = 4.463, \qquad \|X_0 - X_2\| = \sqrt{\frac{1}{100^2} + \frac{1}{100^2}}.$$

As a result, an estimate of the number of iterations required to reach $X_\star$ from X_0 is

$$2\frac{\|X_0 - X_\star\|}{\|X_0 - X_2\|} \approx 631.$$

Note that we have used the information that the norm of the gradient is on average equal to one and that the points X_k, X_{k+1} and X_{k+2} are the vertices of a rectangular triangle with catheti having length $1/100$.

[2 marks]

Answers

2. a) The Lagrangian for the problem is

$$L(x,y,\lambda) = x + 2y + \lambda(y^2 + xy - c^2).$$

The necessary conditions of optimality are

$$0 = \frac{\partial L}{\partial x} = \lambda y + 1, \qquad 0 = \frac{\partial L}{\partial y} = 2 + \lambda(x + 2y), \qquad y^2 + xy - c^2 = 0.$$

One could solve the first two equations to determine x and y as a function of λ and then use the last equation to determine λ. This procedure (or any other alternative solution procedure) yields two candidate optimal solutions

$$S_1^\star = (x_1^\star, y_1^\star, \lambda_1^\star) = (0, -c, 1/c), \qquad S_2^\star = (x_2^\star, y_2^\star, \lambda_2^\star) = (0, c, -1/c).$$

[2 marks]

b) Note that

$$\nabla^2 L = \begin{bmatrix} 0 & \lambda \\ \lambda & 2\lambda \end{bmatrix}$$

and that

$$\frac{\partial g}{\partial (x,y)} = \begin{bmatrix} y & x + 2y \end{bmatrix}.$$

Thus, $\dfrac{\partial g}{\partial (x,y)} s = 0$ yields

$$s = \begin{bmatrix} -x - 2y \\ y \end{bmatrix}.$$

As a result, at $S_1^\star$ the sufficient condition of optimality yields

$$\begin{bmatrix} -2c & c \end{bmatrix} \begin{bmatrix} 0 & 1/c \\ 1/c & 2/c \end{bmatrix} \begin{bmatrix} -2c \\ c \end{bmatrix} = -2c < 0,$$

whereas at $S_2^\star$ one has

$$\begin{bmatrix} 2c & -c \end{bmatrix} \begin{bmatrix} 0 & -1/c \\ -1/c & -2/c \end{bmatrix} \begin{bmatrix} 2c \\ -c \end{bmatrix} = 2c > 0.$$

Therefore $S_1^\star$ identifies a local (strict) maximizer and $S_2^\star$ identifies a local (strict) minimizer.

[4 marks]

c) The values of the objective function at the local maximizer is $f_{\max} = -2c$ and the value at the local minimizer is $f_{\min} = 2c$. Clearly, since $c > 0$, $f_{\min} > f_{\max}$. This is possible if the problem is

- either not bounded from above and/or below, that is if there is a global minimizer which equals $-\infty$ and/or a global maximizer which equals $+\infty$;
- or the global minimum/maximum is achieved as $\|(x,y)\| \to \infty$.

[6 marks]

d) A direct computation shows that

$$\frac{df_{\min}}{dc} = 2, \qquad \frac{\partial L}{\partial c} = -2c\lambda_2^\star = 2,$$

which proves the claim.

[2 marks]

(Note: this is an application of a general result known as the Envelope Theorem, which relates the total derivative of the optimal cost with respect to a parameter to the partial derivative of the optimal Lagrangian with respect to the same parameter.)

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e) Solving the constrain equation for x yields

$$x = \frac{c^2}{y} - y,$$

which replaced in the cost function yields

$$f(x(y),y) = \frac{c^2+y^2}{y} = \frac{c^2}{y} + y.$$

The graph of the function $f(x(y),y)$ is in the figure below. From the graph we observe that the function has a local minimizer and a local maximizer. The value of the function at the local minimizer is above the value of the function at the local maximizer, consistently with what determined in part c). Finally, the function is unbounded from above/below and this is why the local maximizer/minimizer does not solve the optimization problem. [4 marks]

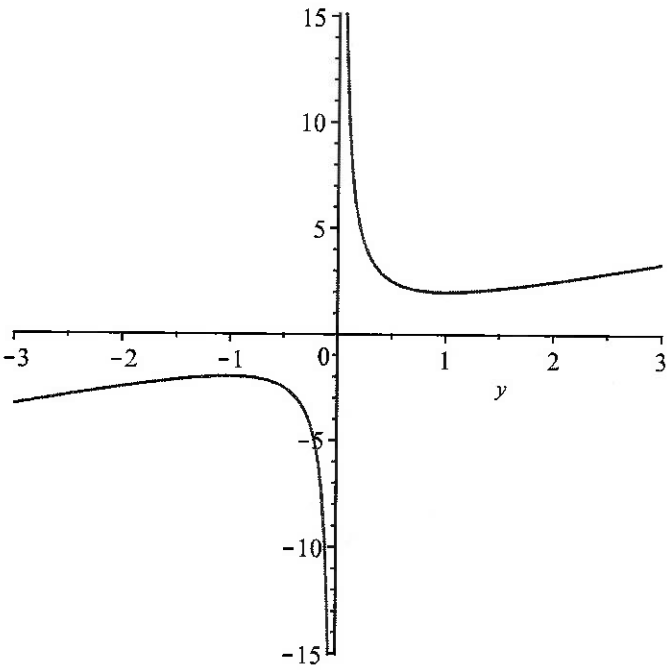
f) For $c = 0$ the necessary conditions of optimality yield

$$\lambda y + 1 = 0, \qquad 2 + \lambda(x+2y) = 0, \qquad y(y+x) = 0.$$

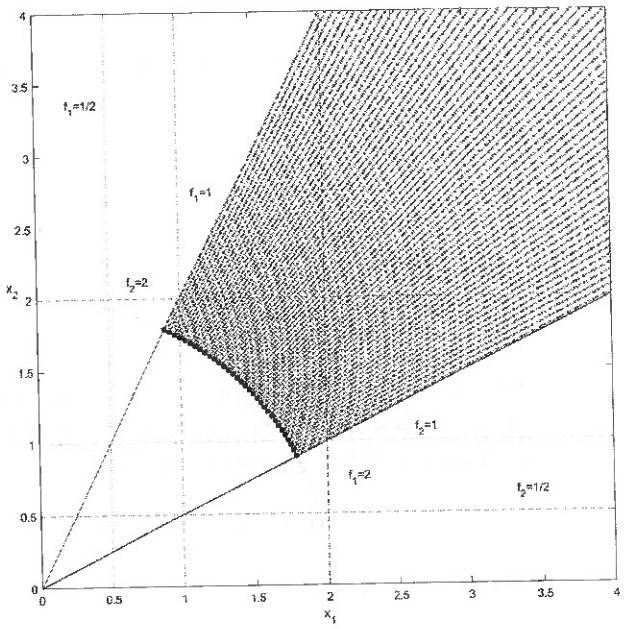
These equations have no solution, hence there is no candidate optimal solution. Finally

$$\lim_{c\to 0} f(x(y),y) = y,$$

which does not have any stationary point. [2 marks]



3. a) The admissible set and the level lines of the objective functions are plotted in the graph below. [2 marks]



b) All Pareto optimal solutions are the points on the portion of circle indicated with dark solid markers. [4 marks]

c) i) The Lagrangian for the problem is

$$L(x_1,x_2)=w_1x_1+w_2x_2+\rho_1(4-x_1^2-x_2^2)+\rho_2(x_2-2x_1)+\rho_3(x_1-2x_2).$$

The first order necessary conditions of optimality are

$$0=\frac{\partial L}{\partial x_1}=w_1-2\rho_1x_1-2\rho_2+\rho_3, \qquad 0=\frac{\partial L}{\partial x_2}=w_2-2\rho_1x_2+\rho_2-2\rho_3,$$

$$4-x_1^2-x_2^2\le 0, \qquad x_2-2x_1\le 0, \qquad x_1-2x_2\le 0,$$

$$\rho_1\ge 0, \qquad \rho_2\ge 0, \qquad \rho_3\ge 0,$$

$$\rho_1(4-x_1^2-x_2^2)=0, \qquad \rho_2(x_2-2x_1)=0, \qquad \rho_3(x_1-2x_2)=0.$$

[4 marks]

ii) Using the complementarity conditions one has eight cases (note that some of these are trivially impossible since, for example, the three constrains can not be active simultaneously).

Case 0: $\rho_1=0,\ \rho_2=0,\ \rho_3=0$. The necessary conditions yield $w_1=0$ and $w_2=0$, which does not give any solution.

Case 1: $\rho_1=0,\ \rho_2=0,\ \rho_3>0$. The necessary conditions yield $w_1+\rho_3=0$, $w_2-2\rho_3=0$ and $x_1-2x_2=0$, which does not give any solution.

Case 2: $\rho_1=0,\ \rho_2>0,\ \rho_3=0$. The necessary conditions yield $w_1-2\rho_2=0$, $w_2+\rho_2=0$ and $x_2-2x_1=0$, which does not give any solution.

Case 3: $\rho_1 = 0$, $\rho_2 > 0$, $\rho_3 > 0$. The necessary conditions yield $w_1 - 2\rho_2 + \rho_3 = 0$, $w_2 + \rho_2 - 2\rho_3 = 0$, $x_2 - 2x_1 = 0$, and $x_1 - 2x_2 = 0$, which does not give any solution.

Case 4: $\rho_1 > 0$, $\rho_2 = 0$, $\rho_3 = 0$. The necessary conditions yield $w_1 - 2x_1\rho_1 = 0$, $w_2 - 2x_2\rho_1 = 0$ and $4 - x_1^2 - x_2^2 = 0$. These yield $x_1 = \frac{w_1}{2\rho_1}$ and $x_2 = \frac{w_2}{2\rho_1}$, which placed into the active constraint and solved for ρ_1 (recall that $\rho_1 > 0$) yields

$$\rho_1 = \frac{\sqrt{w_1^2 + w_2^2}}{4},$$

hence

$$x_1 = \frac{2w_1}{\sqrt{w_1^2 + w_2^2}}, \qquad x_2 = \frac{2w_2}{\sqrt{w_1^2 + w_2^2}}.$$

Note that these points span, for w_1 and w_2 positive, the quarter of circle identified by the boundary of the first constraint. These points are therefore admissible provided

$$2w_2 - 4w_1 < 0, \qquad 2w_1 - 4w_2 < 0,$$

which can be rewritten as

$$0 < w_1 < 2w_2 < 4w_1.$$

In summary, if w_1 and w_2 do not satisfy the above inequalities, the point is not feasible, otherwise all points indicated in the figure with dark marks (with the exception of the points at the boundary of such a line) are candidate points.

Case 5: $\rho_1 >, \rho_2 = 0, \rho_3 > 0$. The necessary conditions yield the candidate point

$$x_1 = \frac{4\sqrt{5}}{5}, \qquad x_2 = \frac{2\sqrt{5}}{5},$$

with multipliers

$$\rho_1 = \frac{\sqrt{5}}{20}(2w_1 + w_2), \qquad \rho_3 = -\frac{1}{5}(w_1 - 2w_2),$$

which is therefore a candidate point provided

$$0 < w_1 \leq 2w_2.$$

Case 6: $\rho_1 >, \rho_2 > 0, \rho_3 = 0$. The necessary conditions yield the candidate point

$$x_1 = \frac{2\sqrt{5}}{5}, \qquad x_2 = \frac{4\sqrt{5}}{5},$$

with multipliers

$$\rho_1 = \frac{\sqrt{5}}{20}(2w_2 + w_1), \qquad \rho_2 = \frac{1}{5}(2w_1 - w_2),$$

which is therefore a candidate point provided

$$0 < w_2 \leq 2w_1.$$

Answers

Case 7: $\rho_1 >, \rho_2 > 0, \rho_3 > 0$. In this case all constraints have to be active, which is clearly impossible.

In summary, if w_1 and w_2 are such that

$$0 < w_1 \leq 2w_2 \leq 4w_1$$

then all Pareto points are candidate optimal solutions of the weighted problem (and vice versa), whereas if the above condition does not hold some (most!) of the Pareto points cannot be recovered solving the weighted problem. [10 marks]

Answers

4. a) A candidate optimal solution for the original unconstrained problem is a point $x^\star$ such that

$$\frac{\partial f_1}{\partial x}(x^\star) + \frac{\partial f_2}{\partial x}(x^\star) = 0,$$

that is $x^\star$ is a stationary point of $f_1 + f_2$.

To find candidate optimal solutions of the distributed problem define the Lagrangian

$$L(x_1, x_2, z, \lambda_1, \lambda_2) = f_1(x_1) + f_2(x_2) + \lambda_1(x_1 - z) + \lambda_2(x_2 - z).$$

The resulting necessary conditions of optimality, yielding candidate optimal solutions for the distributed problem, are

$$0 = \frac{\partial L}{\partial x_1} = \frac{\partial f_1}{\partial x_1}(x_1) + \lambda_1, \qquad 0 = \frac{\partial L}{\partial x_2} = \frac{\partial f_2}{\partial x_2}(x_2) + \lambda_2, \qquad 0 = \frac{\partial L}{\partial z} = -(\lambda_1 + \lambda_2),$$

$$x_1 - z = 0, \qquad x_2 - z = 0.$$

Solving the last two equations yields $x_1 = x_2 = z$, hence

$$\frac{\partial f_1}{\partial x_1}(z) + \frac{\partial f_2}{\partial x_2}(z) = 0,$$

where we have used the condition that $\lambda_1 + \lambda_2 = 0$. As a result, any candidate optimal solution of the distributed problem is also a stationary point of the unconstrained problem and vice versa. [4 marks]

(Note: the main reason to formulate the distributed problem is that the minimization of f_1 and f_2, separately, may be simpler than the minimization of $f_1 + f_2$. In addition, the two minimization tasks can be performed in parallel.)

b) i) For the specific selection of f_1 and f_2 one has

$$f_1(x) + f_2(x) = x^2 + x + \frac{5}{2}.$$

This is a quadratic function with a global minimizer at $x = -1/2$.

ii) The distributed algorithm, for the specific problem, is

$$\begin{aligned}
x_1^{k+1} &= \frac{1}{\eta+1}\left(\eta z^k - \lambda_1^k + 1\right), \\
x_2^{k+1} &= \frac{1}{\eta+1}\left(\eta z^k - \lambda_2^k - 2\right), \\
z^{k+1} &= \frac{1}{2\eta(\eta+1)}\left(2\eta^2 z^k - \eta + \lambda_1^k + \lambda_2^k\right), \\
\lambda_1^{k+1} &= \frac{1}{2\eta+2}\left(\lambda_1^k - \lambda_2^k + 3\eta\right), \\
\lambda_2^{k+1} &= \frac{1}{2\eta+2}\left(-\lambda_1^k + \lambda_2^k - 3\eta\right),
\end{aligned}$$

[4 marks]

If $\lambda_1^k + \lambda_2^k = 0$, for all k, then

$$z^{k+1} = \frac{1}{2(\eta+1)}\left(2\eta z^k - 1\right).$$

To determine the value z^∞ one has to solve the equation $z^{k+1} = z^k = z^\infty$, that is

$$z^\infty = \frac{1}{2(\eta+1)}(2\eta z^\infty - 1),$$

yielding $z = -1/2$, which coincides with the optimal x determined in part b.i). [4 marks]

Using the indicated value of η and the given initial conditions one has (note that $\lambda_1^k + \lambda_2^k = 0$)

$$x_1^0 = 1, \quad x_2^0 = -2, \quad z^0 = 0, \quad \lambda_1^0 = 0, \quad \lambda_2^0 = 0,$$

$$x_1^1 = 0.0099, \quad x_2^1 = -0.0198, \quad z^1 = -0.0049, \quad \lambda_1^1 = 1.4851, \quad \lambda_2^1 = -1.4851,$$

$$x_1^2 = -0.0097, \quad x_2^2 = -0.0099, \quad z^2 = -0.0098, \quad \lambda_1^2 = 1.4998, \quad \lambda_2^2 = -1.4998.$$

The above figures suggest that, with very high speed of convergence, $x_1 \approx x_2 \approx z$ (actually z is an average of x_1 and x_2). This means that the decentralized algorithm makes $x_1 - x_2$ converge to zero and x_1 and x_2 follow z. Note that z, in turn, converges to the optimal solution of the original problem, and this is how convergence of the distributed algorithm to the optimal solution is achieved. [6 marks]

Answers